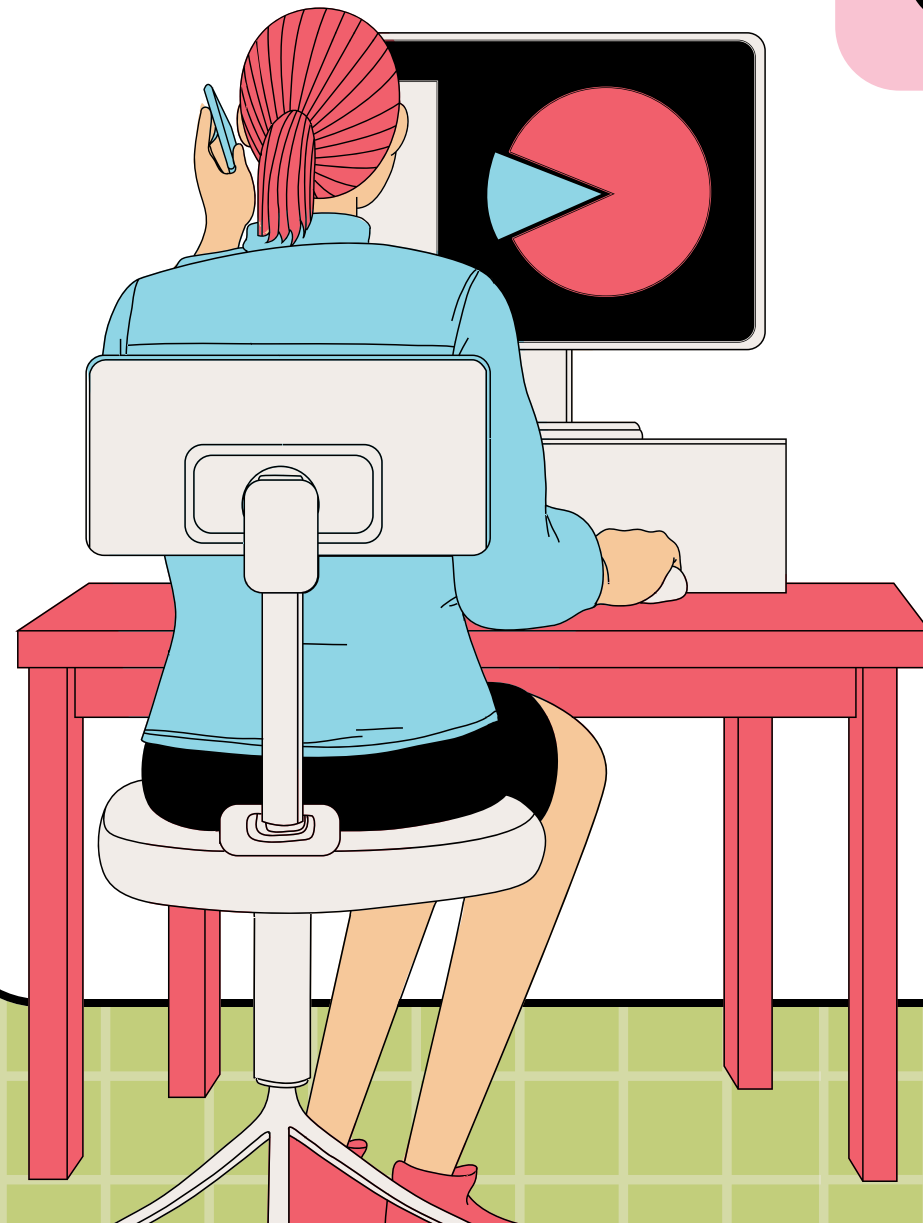
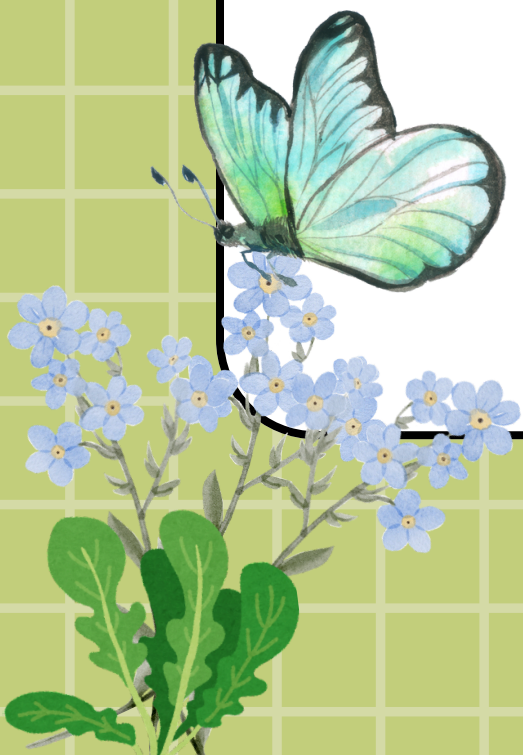


OBESITY LEVEL PREDICTION USING MACHINE LEARNING



PROBLEM STATEMENT

- Obesity is a growing global health issue
- Difficult to manually assess risk level accurately
- Need a data-driven system to classify obesity levels
- **Goal:** Predict obesity category using lifestyle & physical attributes



DATASET DESCRIPTION

16 Features + 1 Target Variable

- **Dataset (Kaggle):** Obesity levels in individuals from the countries of Mexico, Peru and Colombia, based on their eating habits and physical condition.
- **Demographic:** Age, Gender, Height, Weight, Family History
- **Eating Habits:** High-calorie food, Vegetables, Meals, Snacks, Alcohol
- **Lifestyle:** Physical Activity, Water Intake, Smoking, Tech Usage, Transport
- **Target variable:** The obesity level NObeyesdad is the response variable (with 7 different levels),(Underweight, Normal, Overweight, Obesity Type I, II, III)



DATASET OVERVIEW

Shape: (2111, 17)

	Age	Gender	Height	Weight	CALC	FAVC	FCVC	NCP	SCC	SMOKE	CH20	family_history_with_overweight	FAF	TUE	CAEC	MT	
0	21.0	Female	1.62	64.0	no	no	2.0	3.0	no	no	2.0		yes	0.0	1.0	Sometimes	Public_Transpor
1	21.0	Female	1.52	56.0	Sometimes	no	3.0	3.0	yes	yes	3.0		yes	3.0	0.0	Sometimes	Public_Transpor
2	23.0	Male	1.80	77.0	Frequently	no	2.0	3.0	no	no	2.0		yes	2.0	1.0	Sometimes	Public_Transpor
3	27.0	Male	1.80	87.0	Frequently	no	3.0	3.0	no	no	2.0		no	2.0	0.0	Sometimes	Wa
4	22.0	Male	1.78	89.8	Sometimes	no	2.0	1.0	no	no	2.0		no	0.0	0.0	Sometimes	Public_Transpor

Columns: ['Age', 'Gender', 'Height', 'Weight', 'CALC', 'FAVC', 'FCVC', 'NCP', 'SCC', 'SMOKE', 'CH20', 'family_history_with_overweight', 'FAF', 'TUE', 'CAEC', 'MTRANS', 'NObeyesdad']

MTRANS	NObeyesdad
Public_Transportation	Normal_Weight
Public_Transportation	Normal_Weight
Public_Transportation	Normal_Weight
Walking	Overweight_Level_I
Public_Transportation	Overweight_Level_II

verweight', 'FAF', 'TUE', 'CAEC', 'MTRANS', 'NObeyesdad']

ML PIPELINE OVERVIEW AND MATRICS

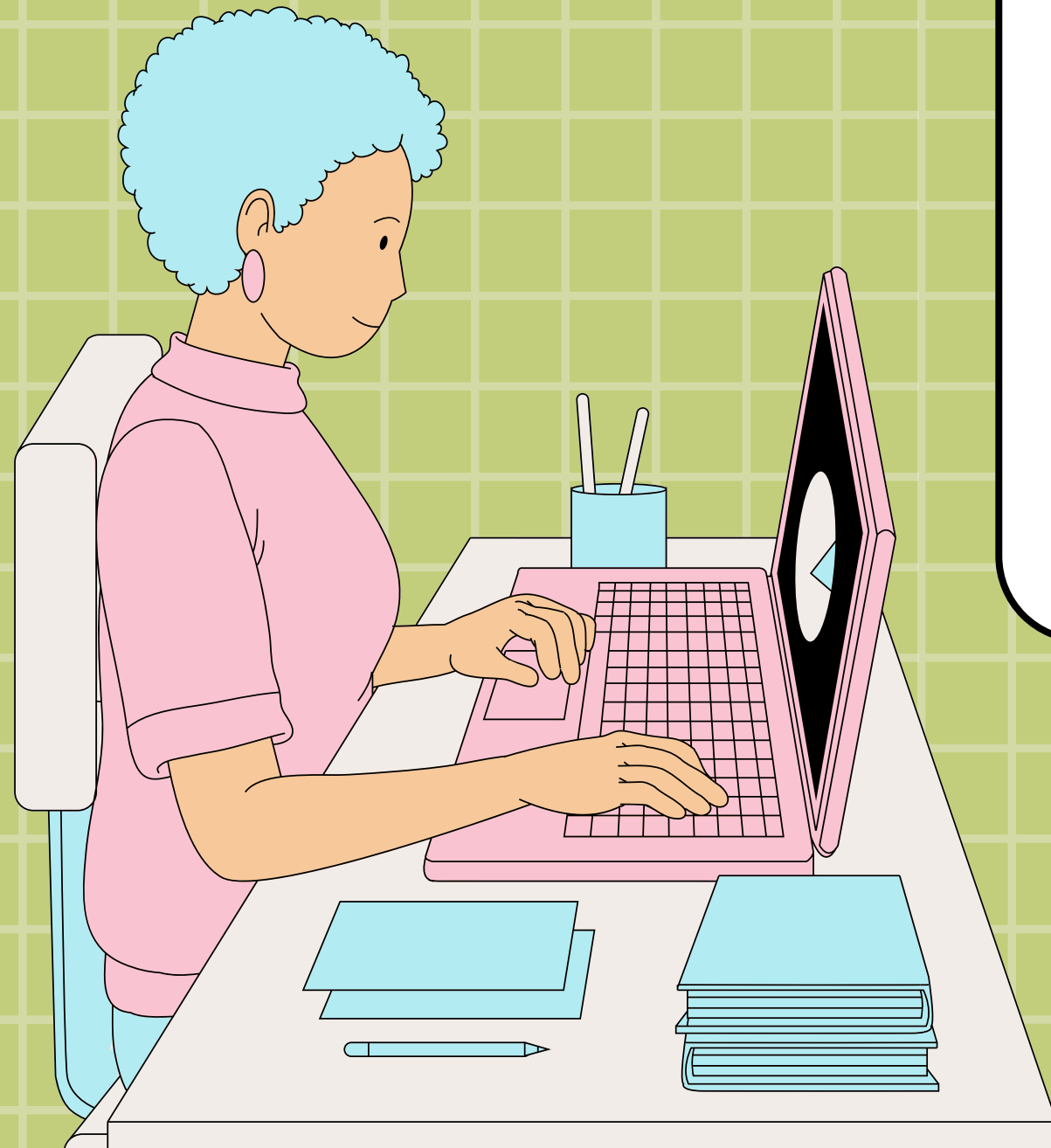
ML pipeline:

- 1.Data Cleaning
- 2.Train-Test Split
- 3.Feature Preprocessing
- 4.Model Training
- 5.Model Evaluation
- 6.Best Model Selection
- 7.Model Saving for Deployment

Metrics Use:

- 1.accuracy
- 2.macro-F1
- 3.confusion matrix

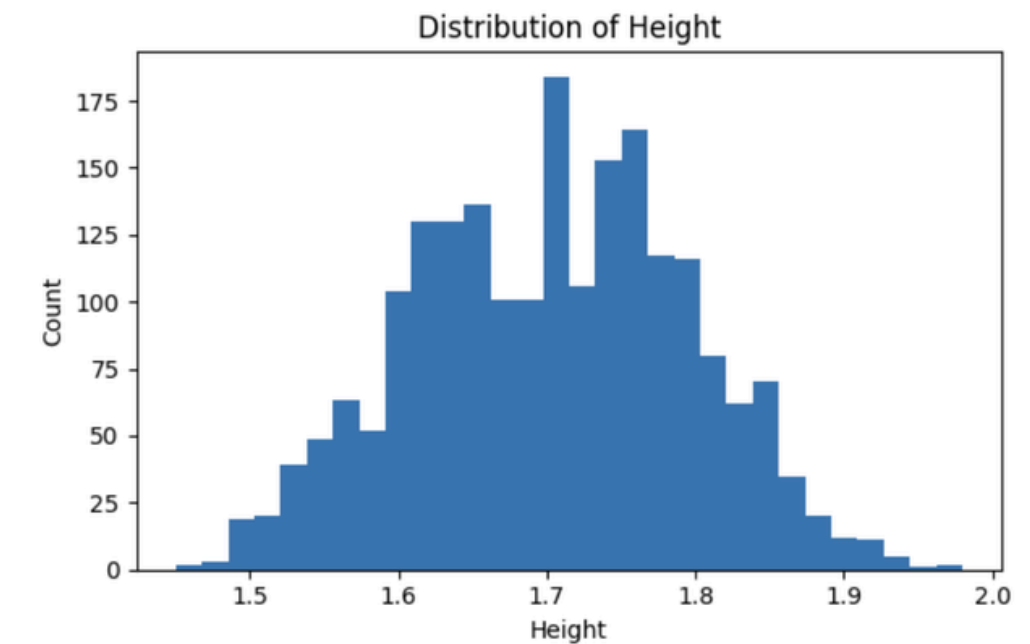
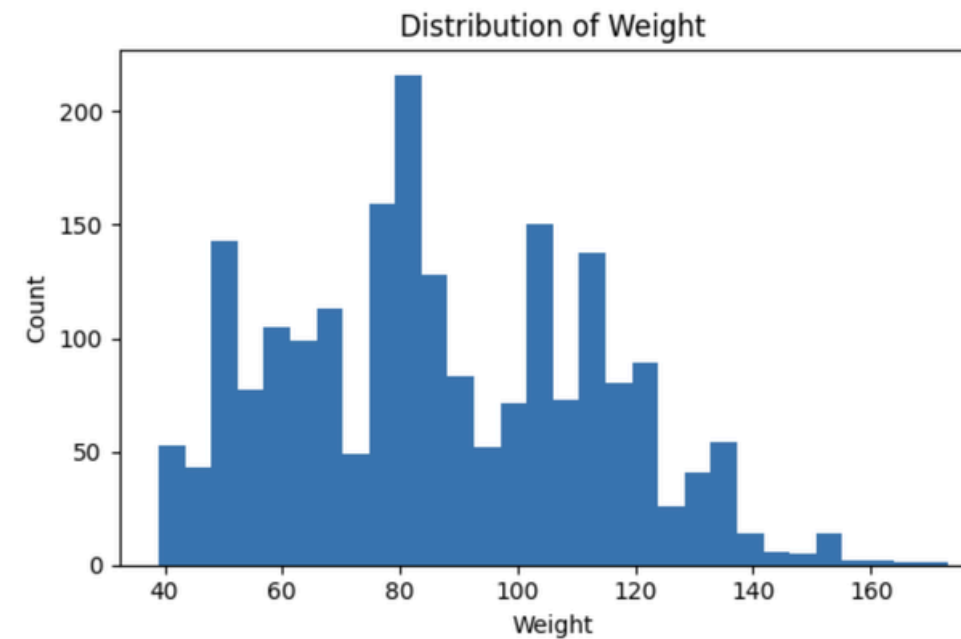
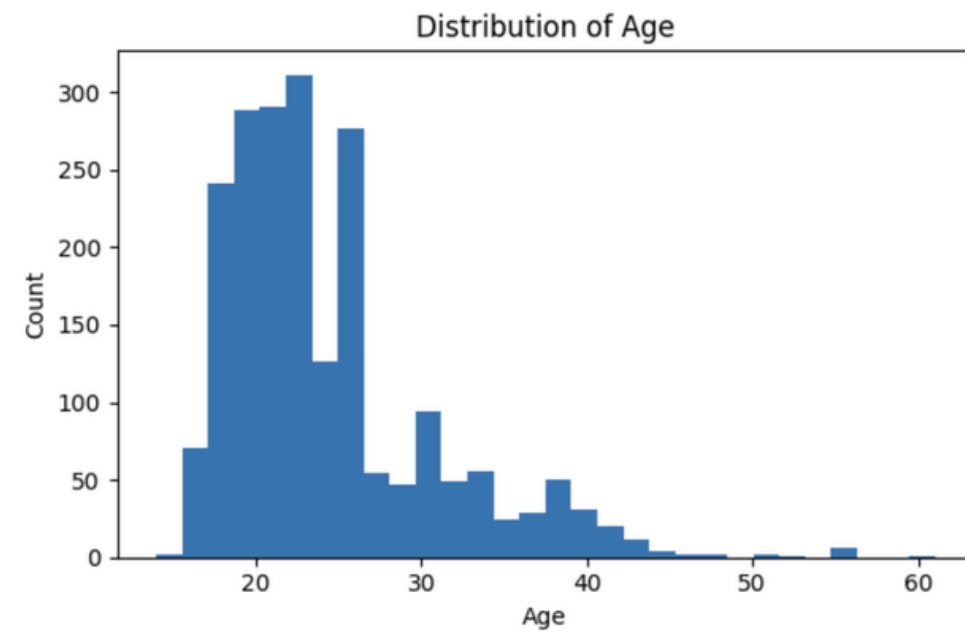




ALGORITHMS USED

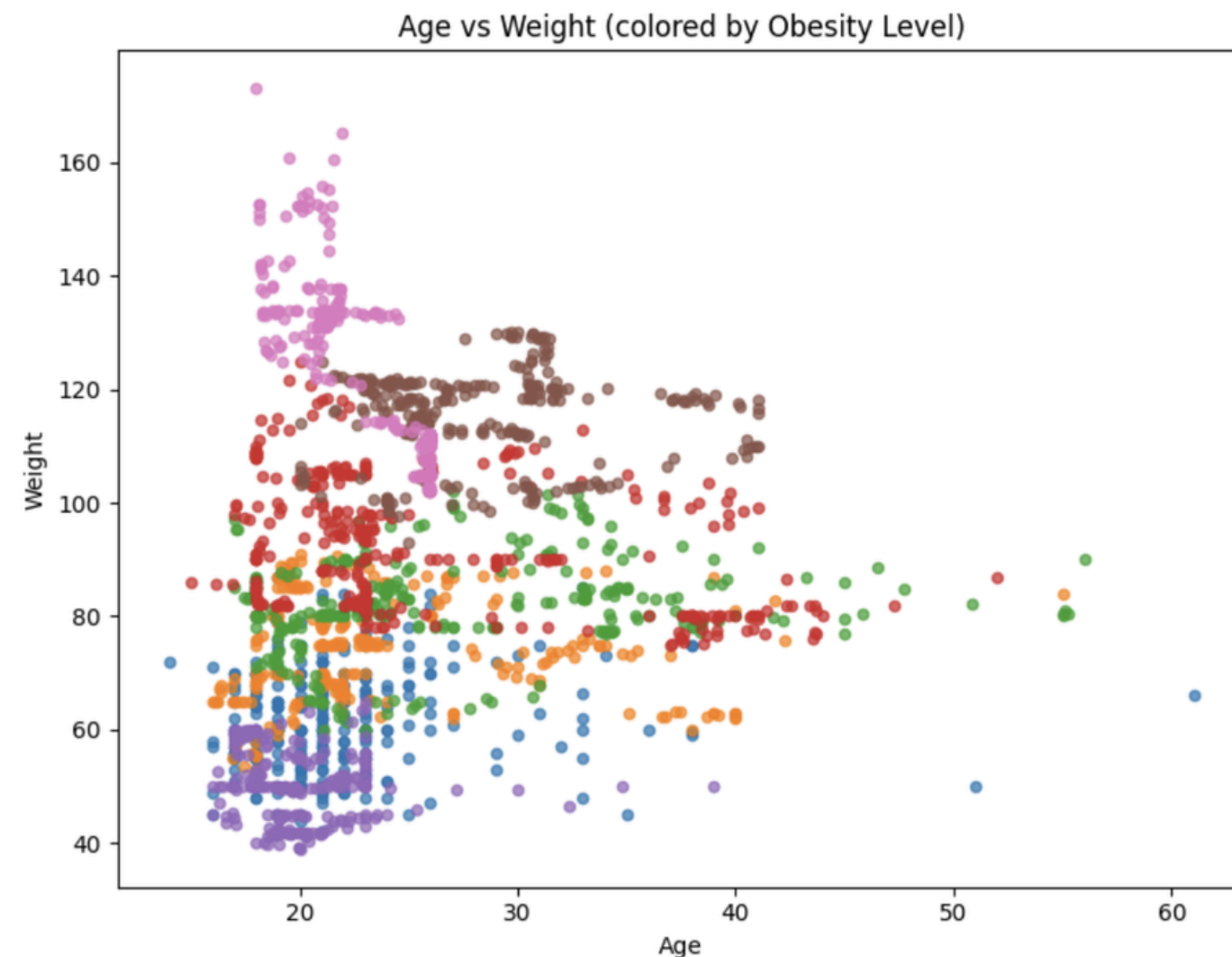
- Random Forest Classifier
- Decision Tree Classifier
- Logistic Regression
- Support Vector Machine (SVM)

EXPLORATORY DATA ANALYSIS



- Majority of participants are 18–30 years old
- Weight shows wide variation meaning strong impact on obesity prediction
- Height follows approximately normal distribution

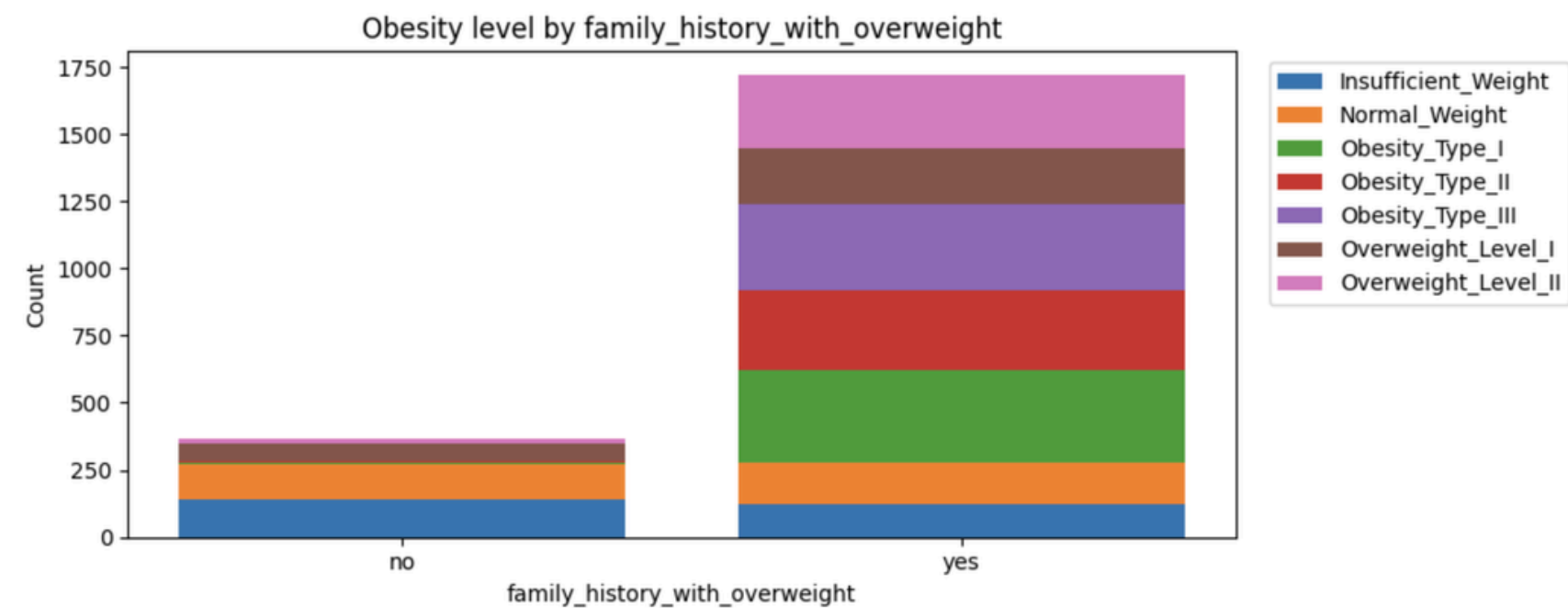
HOW DO AGE AND WEIGHT RELATE TO EACH OTHER?



- Shows relationship between age and weight across obesity levels.
- Higher weight ranges are strongly associated with higher obesity levels.
- Overlapping regions indicate obesity depends on multiple factors, not age alone.
- Younger individuals can still fall into higher obesity levels if weight is high.



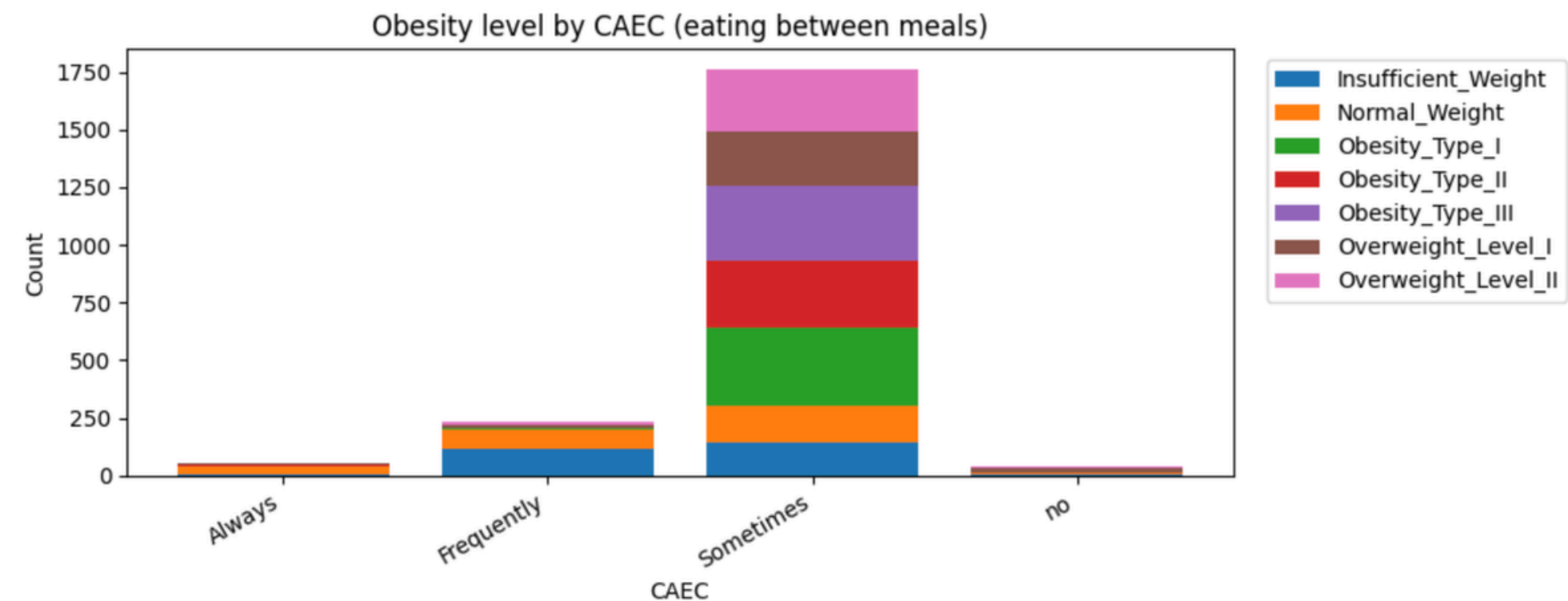
DOES FAMILY HISTORY WITH OVERWEIGHT AFFECT OBESITY TYPE?



- Participants with a family history of overweight show a higher representation in overweight and obesity categories.
- This suggests a possible genetic or shared lifestyle influence on obesity risk.
- Genetic or lifestyle influence may contribute to obesity risk.
- Non-family-history group has more normal-weight individuals.



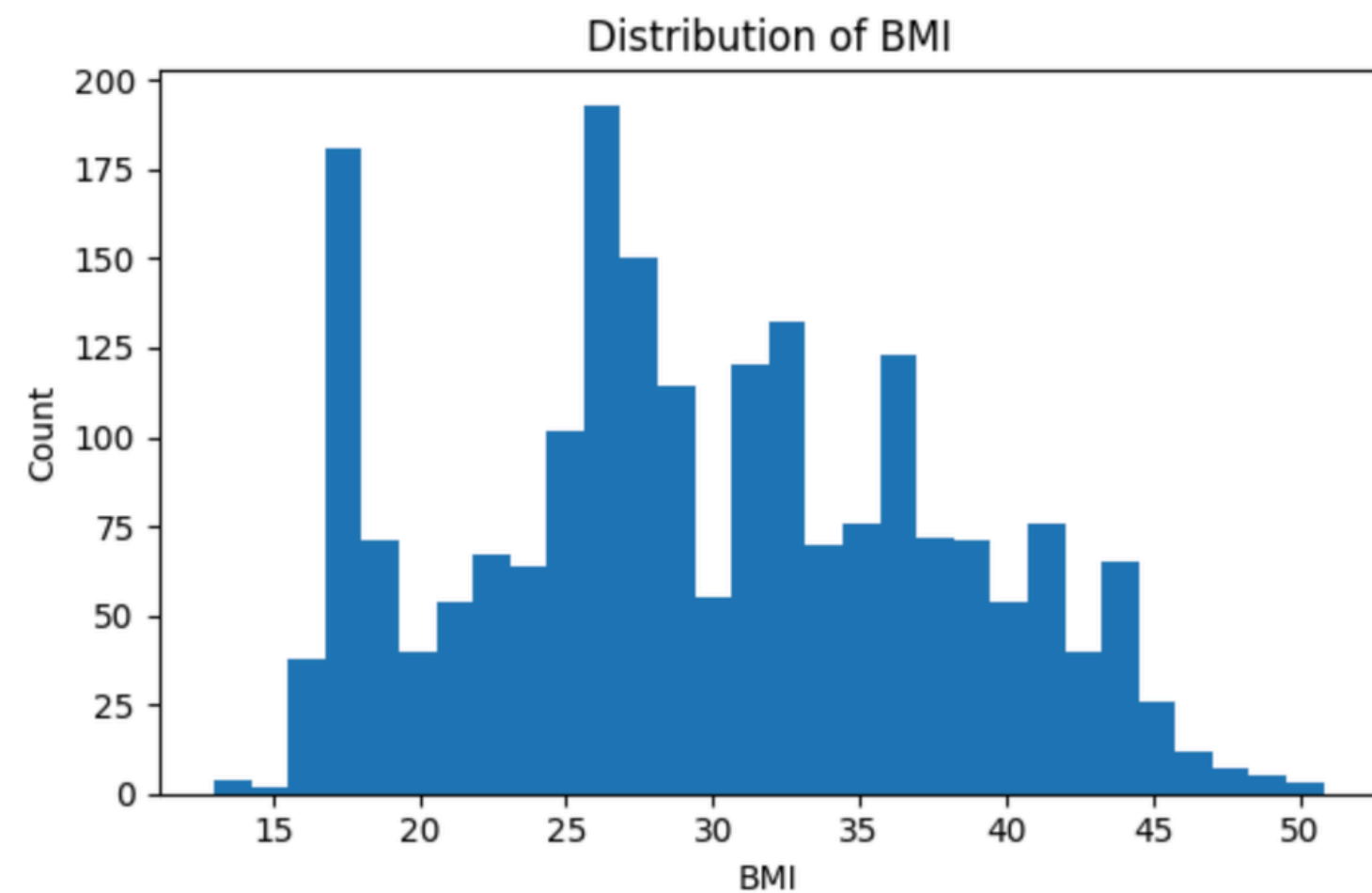
HOW EATING FOOD BETWEEN MEALS AFFECT OBESITY



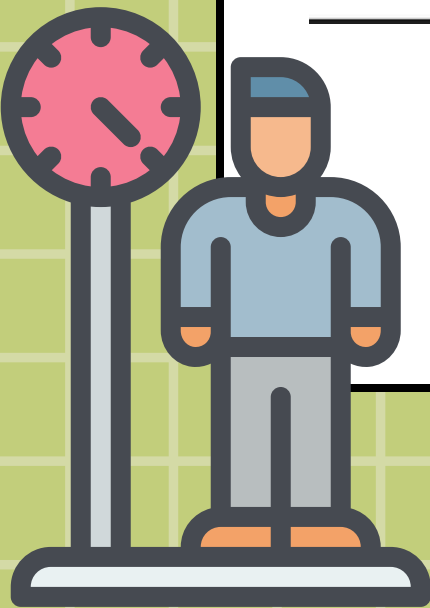
- Obesity appears across all snacking categories, but higher snacking frequency shows greater representation in overweight and obesity classes.
- This indicates that eating behavior may contribute to weight gain patterns.
- Individuals who eat between meals more often tend toward overweight/obese classes.
- Eating habits significantly influence body weight.



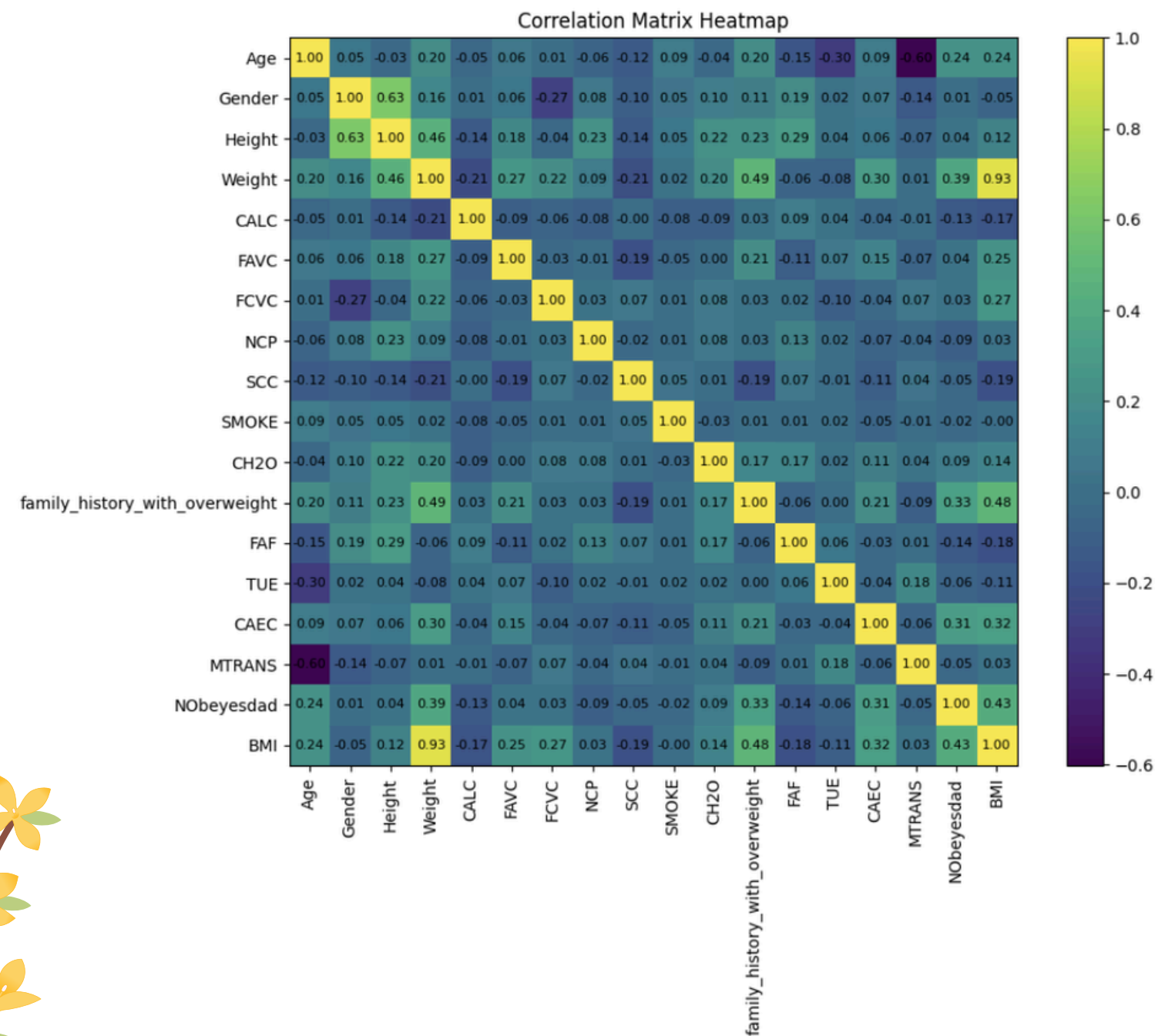
BMI



- Added BMI since there are weight and height
- BMI ranges from about 15 to 50+
- Most values fall between 20 – 35
- Includes normal, overweight, and obese categories
- BMI is a key feature for obesity prediction

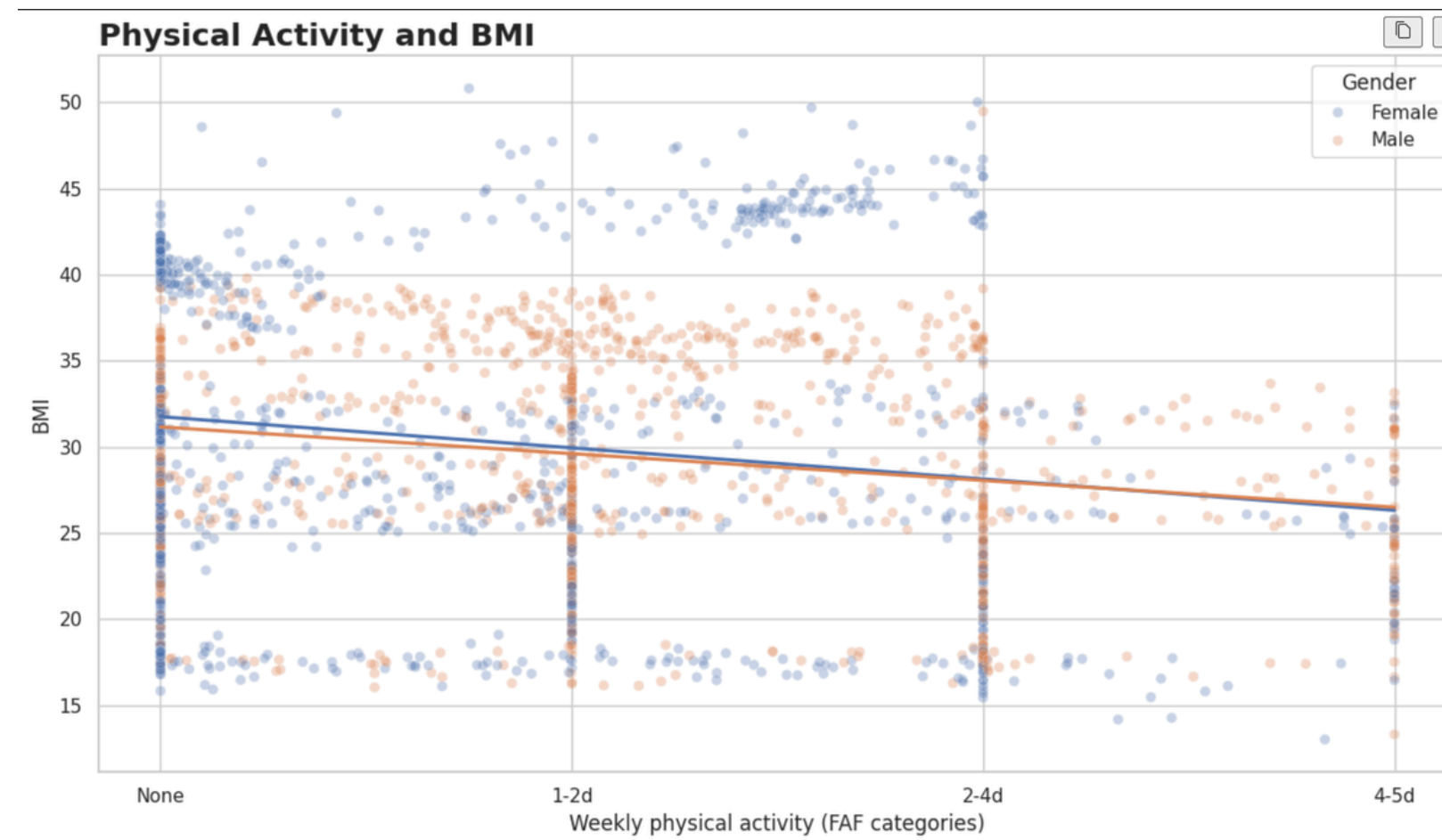


CORRELATION MATRIX HEATMAP (WITH VALUES)



- Displays relationships between all features.
- BMI and weight show strong positive correlation with obesity level, confirming they are strong predictive features.
- Lifestyle variables show weaker correlations, suggesting obesity is multi-factorial.
- Helps to identify important predictors for machine learning models.

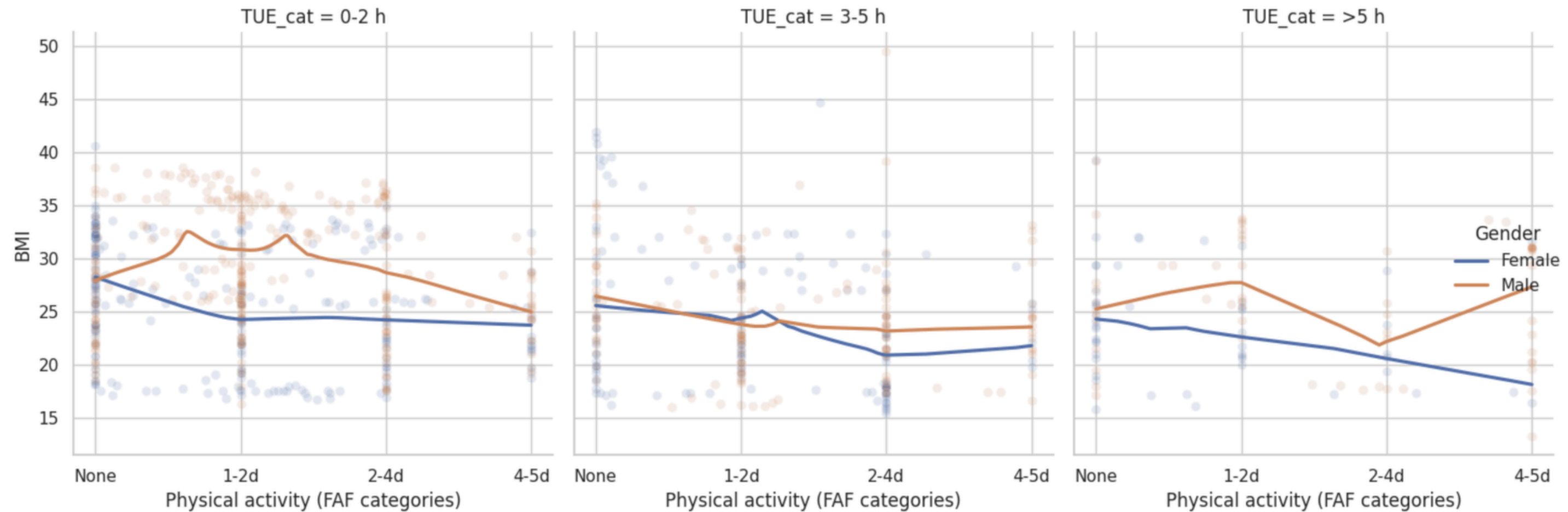
PHYSICAL ACTIVITY AND BMI



- Increased physical activity is associated with slightly lower BMI, but wide variability indicates exercise alone is not sufficient to predict obesity status.
- Similar downward trend for both males and females.
- More frequent exercise links to lower BMI.
- Wide data spread shows exercise alone doesn't determine BMI.
- Other lifestyle factors also affect BMI.



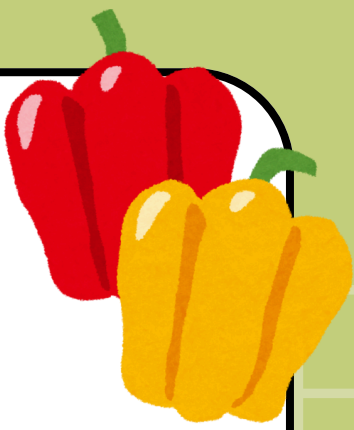
PHYSICAL ACTIVITY VS BMI BY SCREEN TIME



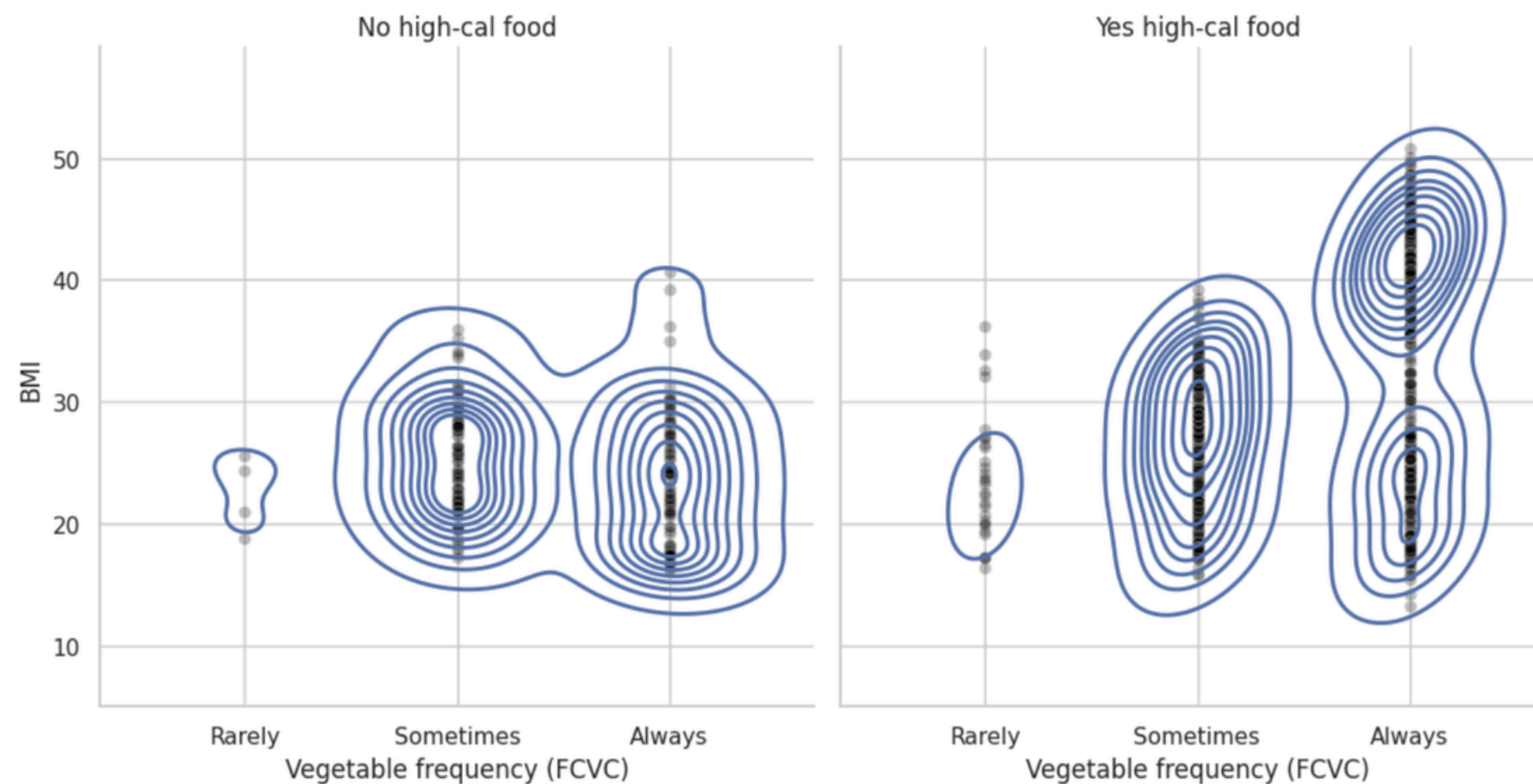
- BMI generally decreases as physical activity increases.
- In 0–2 hour group, BMI slowly goes down with more exercise
- In 3–5 hour group, BMI decreases but slightly increases at the highest activity level..
- High screen time >5 hour group, the pattern is less clear, but higher activity still links to lower BMI.
- more exercise is associated with lower BMI.



BMI VS VEGETABLE FREQUENCY



BMI vs vegetable frequency



Without high-calorie food:

- Higher vegetable intake links to lower BMI.
- “Always” vegetable eaters show the leanest BMI ($\sim 24 \text{ kg/m}^2$).

With high-calorie food:

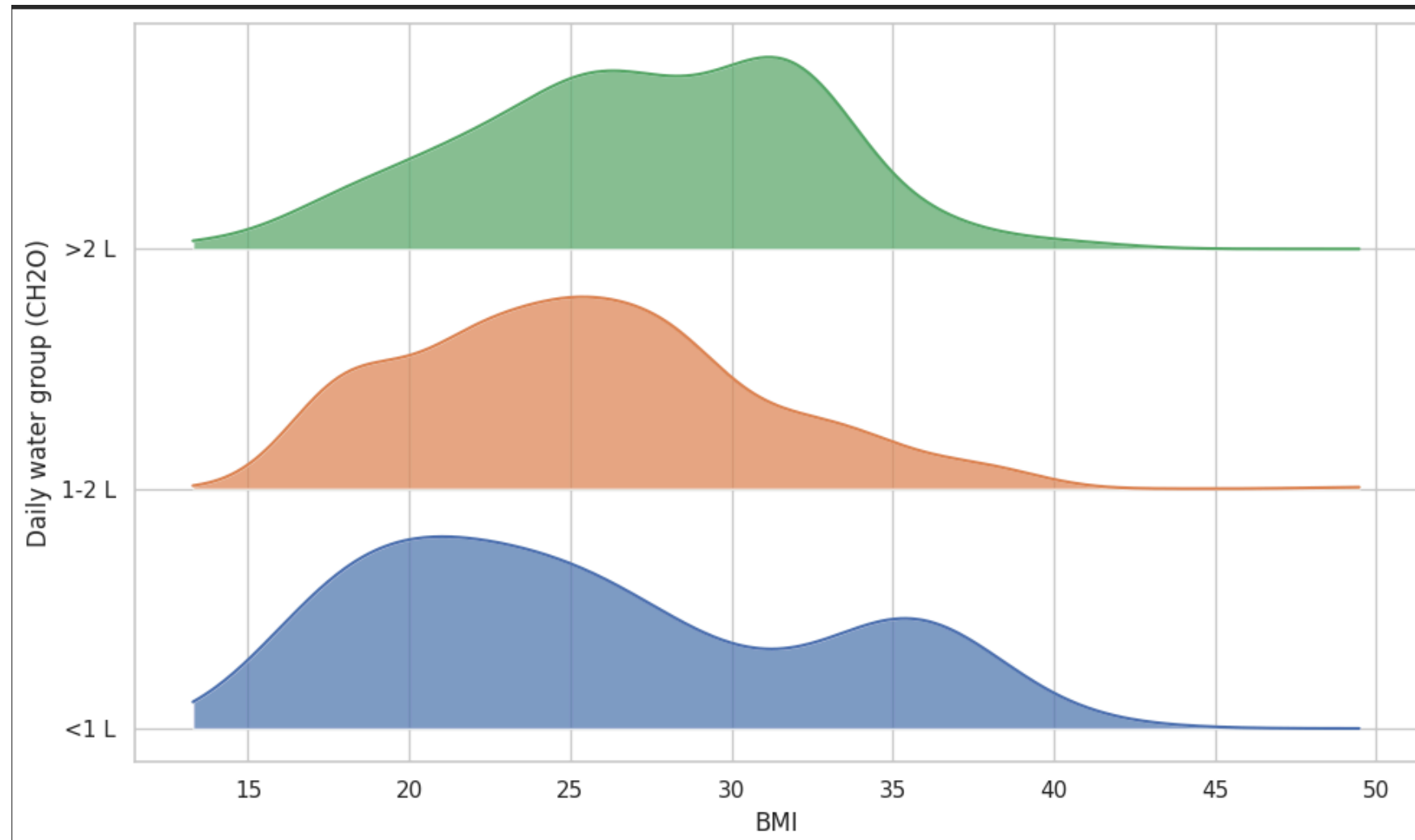
- When high-calorie food is consumed, BMI remains elevated across all vegetable frequency groups.
- Very high BMI values ($\approx 40\text{--}45 \text{ kg/m}^2$) appear even among “Always” vegetable consumers.

Vegetable intake alone is insufficient to offset the impact of high-calorie food consumption.

Overall dietary composition appears to influence BMI more strongly than a single healthy habit.



HYDRATION AND BMI OF AGE GROUP



< 1 L/day:

- Widest BMI distribution.
- Peak around mid-20s but many values above BMI 30.
- Indicates higher obesity variation with low hydration.

1-2 L/day:

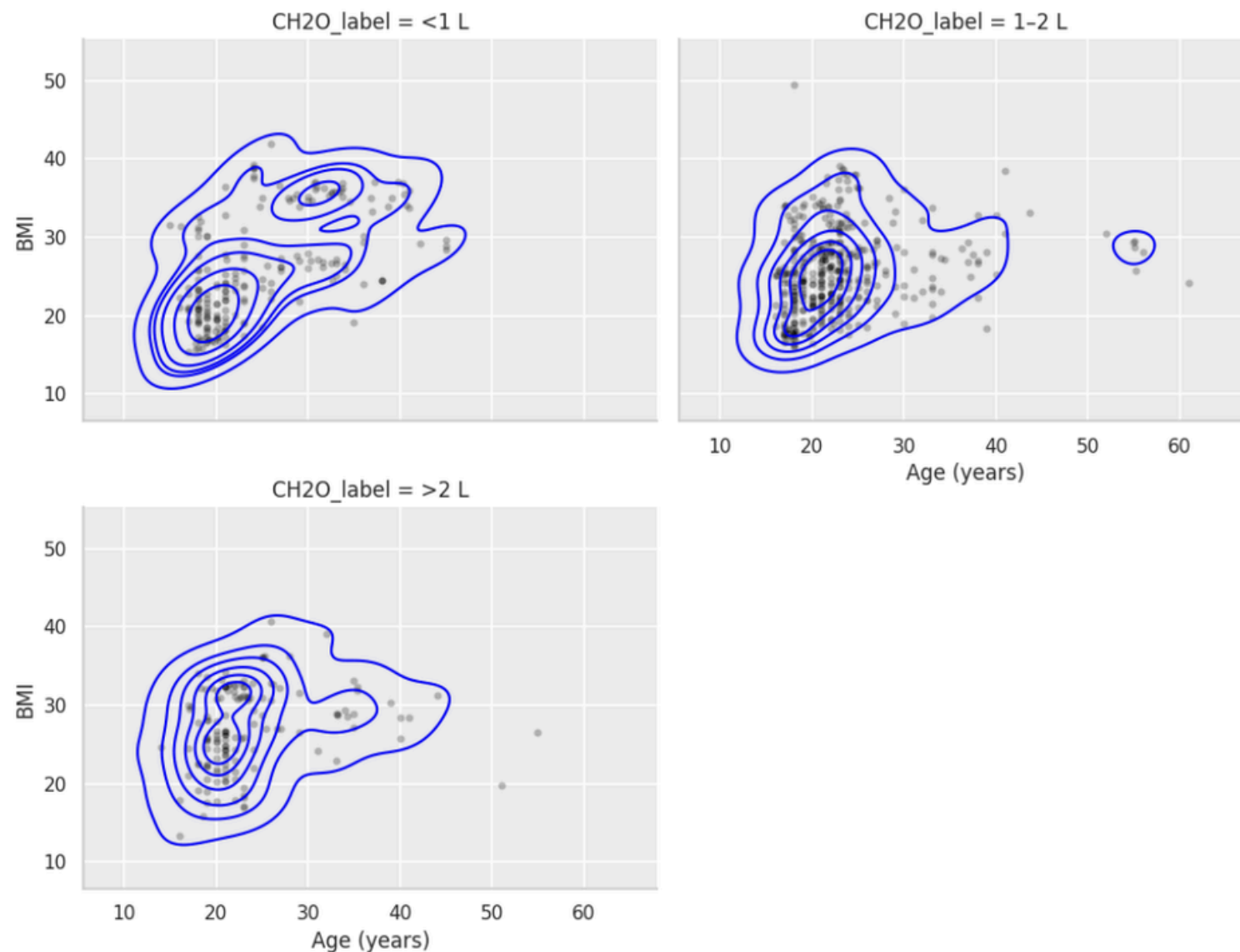
- More concentrated BMI distribution.
- Centered around high-20s.
- Fewer very high BMI values.

> 2 L/day:

- Narrower and slightly lower BMI peak (mid-20s).
 - More consistent BMI levels.
-
- Higher hydration levels show more concentrated BMI distribution, but no strong reduction in obesity level is observed.
 - Water intake alone does not significantly determine BMI.



AGE, HYDRATION, AND BMI



< 1 L group:

- Highest density around age ~18–22
- BMI mostly clustered in the normal range (~20–23), with some higher-BMI individuals

1–2 L group:

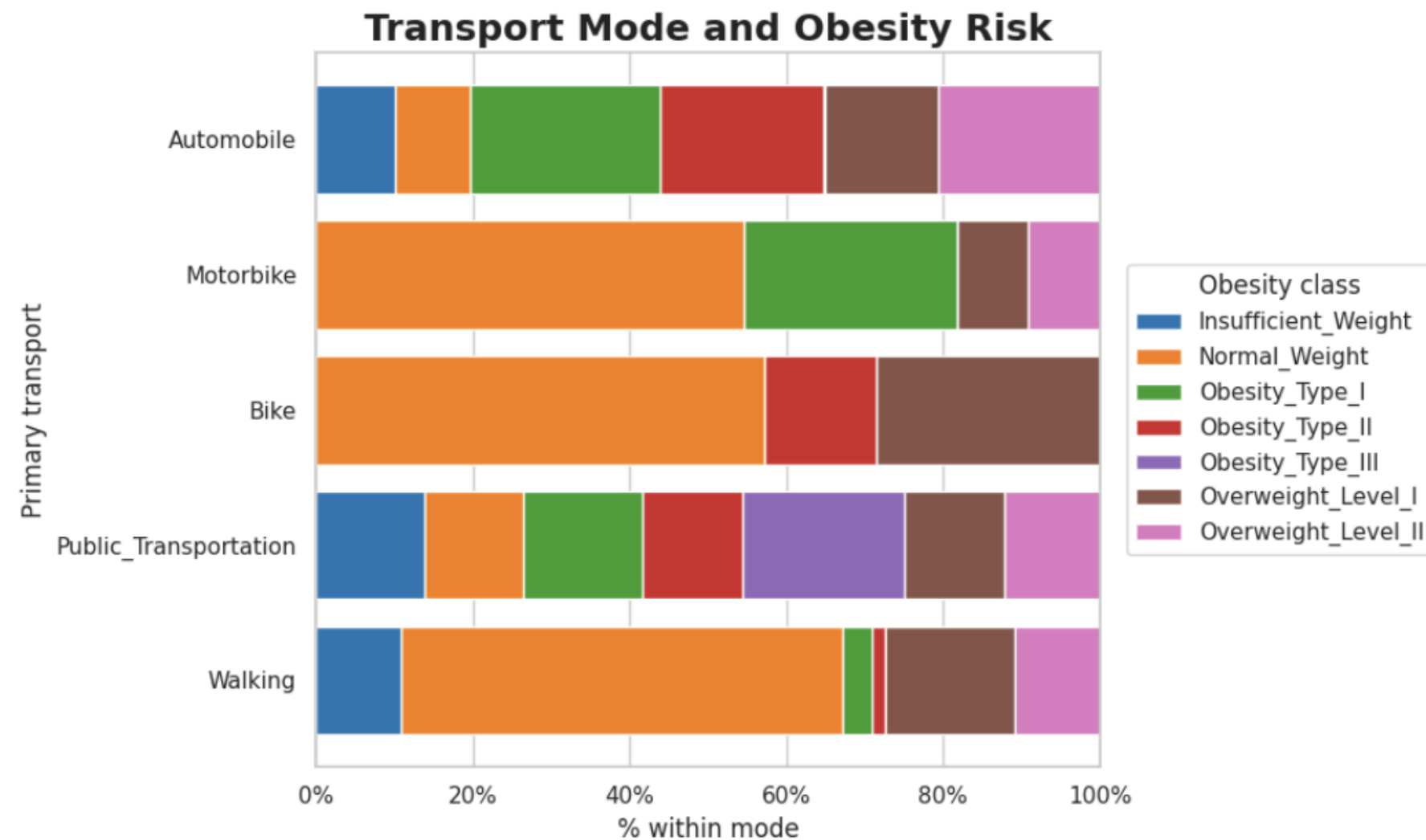
- Participants slightly older (early to mid-20s)
- BMI commonly between ~23–28 (upper normal to overweight range)

> 2 L group:

- BMI concentrated in mid-20s despite higher water intake
- No clear evidence that higher hydration corresponds to lower BMI



TRANSPORT MODE AND OBESITY RISK



Walking:

- Highest percentage of Normal Weight individuals.
- Walking shows the highest proportion of Normal Weight individuals and relatively low obesity representation in this dataset.

Biking:

- Similar to walking but slightly more Overweight Level I.
- Small presence of higher obesity levels.

Public Transportation:

- Most mixed obesity-class distribution.
- Lower Normal Weight percentage.
- Higher share across overweight and obesity categories.

Motorbike:

- Moderate Normal Weight group.
- Noticeable Obesity Type I and overweight levels.

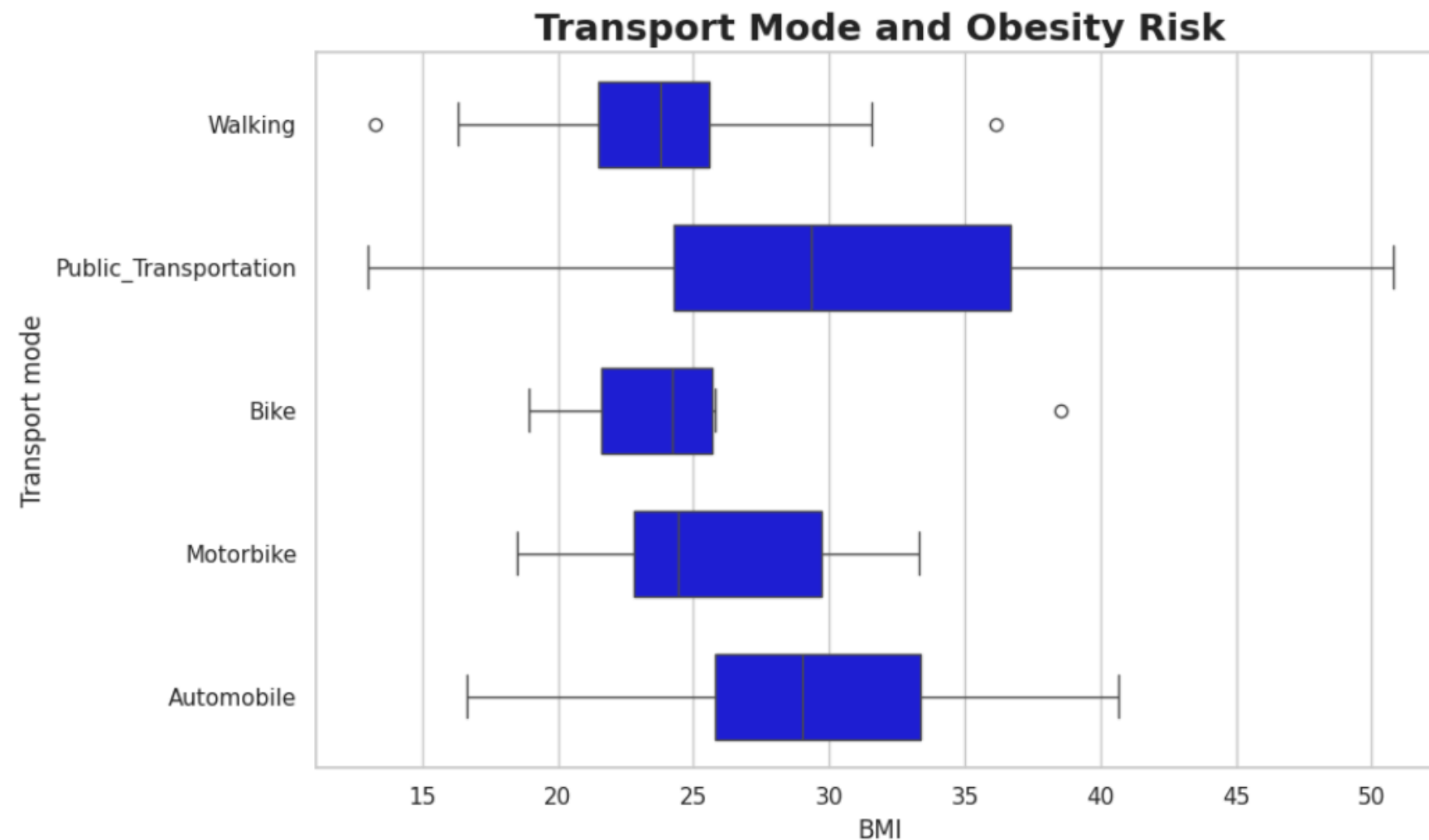
Automobile:

- Highest obesity proportion, especially Obesity Type I & II.
- Smallest Normal Weight share.

- Active transport modes (walking, biking) show higher proportions of normal-weight individuals, while automobile use shows higher obesity representation.
- This suggests mobility behavior may relate to obesity risk patterns.



TRANSPORT MODE AND BMI (BOXPLOT)



Walking:

- Lowest median BMI (~23 kg/m²).
- Narrow spread (tight IQR).
- Very few high-BMI outliers → healthiest profile.

Biking:

- BMI clustered around mid-20s.
- Compact and consistent distribution.

Motorbike:

- Slightly higher median BMI (~24–25 kg/m²)
- Wider variation compared to active transport.

Public Transportation:

- Widest BMI spread.
- Widest BMI spread (~14–50 kg/m²)

Automobile:

- Highest median BMI (~29–30 kg/m²)
- Large IQR indicating greater obesity risk variation.

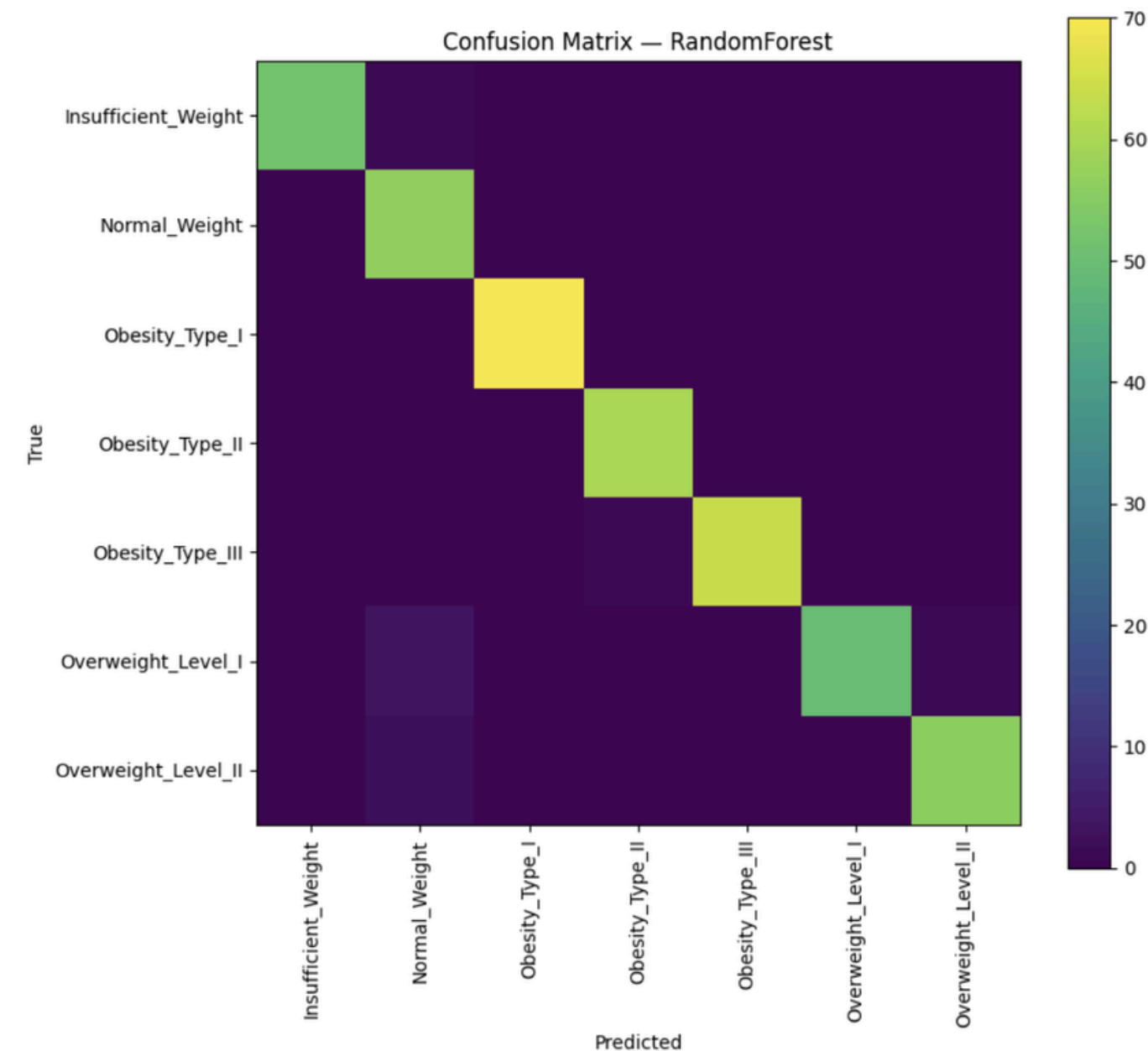


BEST MODEL)

	Model	Accuracy	MacroF1
0	RandomForest	0.978469	0.978150
1	DecisionTree	0.976077	0.975423
2	LogisticRegression	0.921053	0.919198
3	LinearSVM_Calibrated	0.822967	0.814443
Best model: RandomForest			

- **RandomForest**
- Highest performance among all tested models
- Selected as final model

CONFUSION MATRIX



- Most predictions lie on the diagonal > high correct classification
- Very few misclassifications between obesity categories
- Model performs consistently across all 7 classes
- Good class balance (no major bias toward one category)

WEB APPLICATION FEATURES

Obesity Level Prediction

Educational demo. Not medical advice.

Input features

Body Measurements

Height (cm)
170

Weight (lb)
183

BMI (kg/m²): 28.72

BMI is one indicator. The prediction result is based on all inputs.

Auto-calculated from: 183 lb, 170 cm

Other Numeric Features

Age (years)
23

Vegetable consumption frequency (scale 1-3)
2.396

Number of main meals (meals/day)
3.000

Daily water intake (liters/day)
2.000

Prediction result

Predicted obesity level: Overweight Level II

Top-1 probability: 0.813

Top results

Class	Probability
Overweight Level II	0.8133
Overweight Level I	0.1233
Obesity Type I	0.0533
Normal Weight	0.0067
Obesity Type II	0.0033

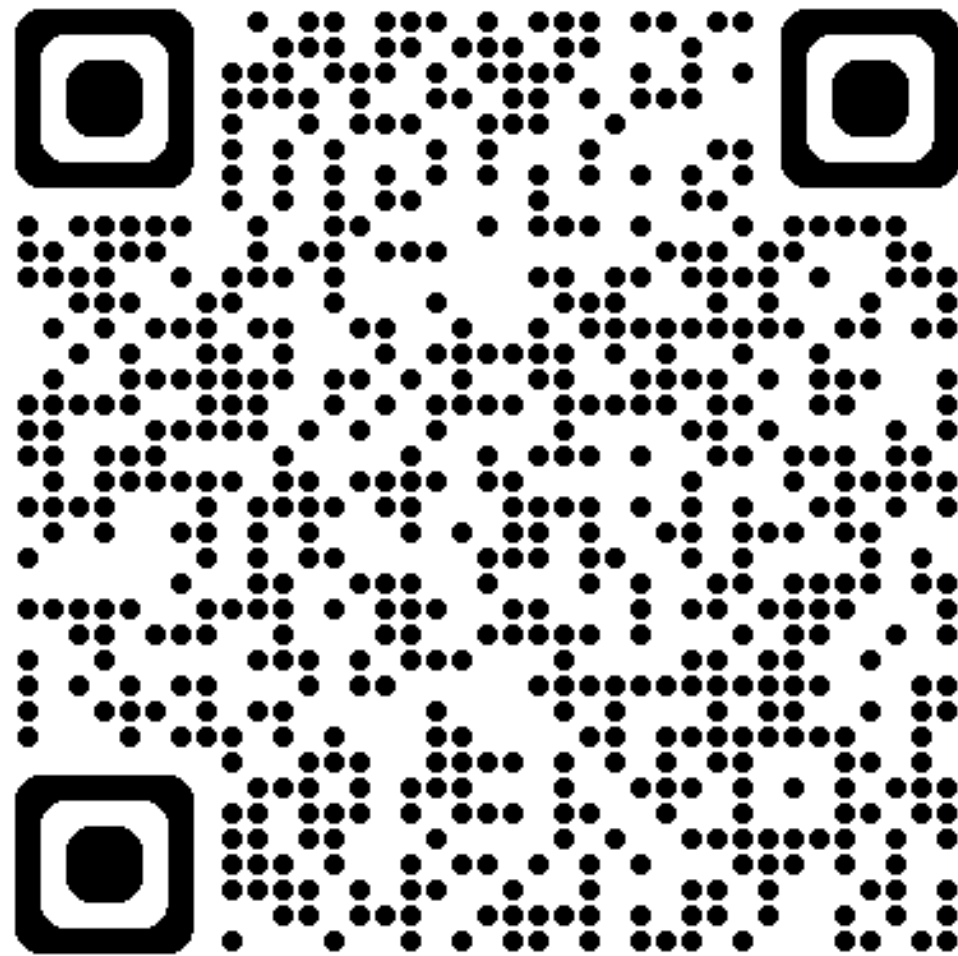
Probability chart (Top-K)

Top-K Class Probabilities

- User enters health & lifestyle data
- System preprocesses input automatically
- Model predicts obesity level instantly
- User-friendly interface



WEB APPLICATION



SCAN QR CODE

Obesity Level Predictor



THANK YOU

